

Leibnitz Theorem

If u and v are two functions, each differentiable n times at a common point in their domain, then the n -th derivative of their product at that point is given by

$$(uv)_n = \sum_{r=0}^n \binom{n}{r} u_{n-r} v_r = u_n v + \binom{n}{1} u_{n-1} v_1 + \binom{n}{2} u_{n-2} v_2 + \cdots + \binom{n}{r} u_{n-r} v_r + \cdots + u v_n,$$

where suffixes denote the order of derivative at the common point of the functions u , v , and $u_0 = u$, $v_0 = v$.

OR

Statement : If u and v are two functions of x , possessing derivatives of the n^{th} order, then

$$(uv)_n = {}^n C_0 u_n v + {}^n C_1 u_{n-1} v_1 + {}^n C_2 u_{n-2} v_2 + \dots + {}^n C_r u_{n-r} v_r + \dots + {}^n C_n u v_n.$$

OR

If u and v are any two functions of x such that all their desired differential coefficients exist, then the n^{th} differential coefficient of their product is given by

$$D^n(uv) = (D^n u) \cdot v + {}^n C_1 D^{n-1} u \cdot Dv + {}^n C_2 D^{n-2} u \cdot D^2 v + \dots + {}^n C_r D^{n-r} u \cdot D^r v + \dots + u D^n v.$$

Q.

$$y = x^4 \cos x, \quad x \in \mathbb{R}.$$

Use the Leibniz rule to find a simplified expression for $\frac{d^6 y}{dx^6}$.

Example 14 : Find y_n if $y = x^2 \log x$,

3. (Application) Find the n^{th} derivative of (a) $x^3 \log x$ **Ans:** (a) $(-1)^n (n-4)! 6 x^{-n+3}$;

(b) $\tan^{-1} \left(\frac{2x}{1-x^2} \right)$, **Ans:** $(-1)^{n-1} (n-1)! \sin^n \theta \cdot \sin n\theta$;

(c) $\cos^2 x \sin^3 x$ **Ans:** $\frac{1}{16} \{ 2 \sin \left(x + n \frac{\pi}{2} \right) + 3^n \sin \left(3x + n \frac{\pi}{2} \right) - 5^n \sin \left(5x + n \frac{\pi}{2} \right) \}$;

(d) $e^x \sin 4x \cos 6x$, **Ans:** $e^x \frac{(101)^{\frac{n}{2}}}{2} \sin(10x + n \tan^{-1} 10) - e^x \frac{(5)^{\frac{n}{2}}}{2} \sin(2x + n \tan^{-1} 10)$;

Example 13 : Find y_n if $y = x^2 e^{ax}$.

Solution : By Leibnitz's Theorem, if $y = uv$, then

$$y_n = u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \dots + {}^nC_n u v_n \quad \dots(1)$$

We are given that $y = x^2 e^{ax}$.

Therefore, we can take $u = e^{ax}$, $v = x^2$.

Then we have $u_1 = ae^{ax}$, $u_2 = a^2 e^{ax}$, $u_3 = a^3 e^{ax}$,, $u_n = a^n e^{ax}$

and $v_1 = 2x$, $v_2 = 2$, $v_3 = 0$, $v_4 = 0$,, $v_n = 0$.

Substituting these values in equation (1), we get

$$\begin{aligned} y_n &= a^n e^{ax} x^2 + {}^nC_1 a^{n-1} e^{ax} (2x) + {}^nC_2 a^{n-2} e^{ax} (2) + 0 + 0 + \dots \\ &= a^n e^{ax} x^2 + n a^{n-1} e^{ax} (2x) + \frac{n(n-1)}{2} a^{n-2} e^{ax} (2) \\ &= a^{n-2} e^{ax} [a^2 x^2 + 2nax + n(n-1)]. \end{aligned}$$

Example11 Find the n^{th} derivative of $x \log x$

Solution: Let $u = \log x$ and $v = x$

$$\text{Then } u_n = (-1)^{n-1} \frac{(n-1)!}{x^n} \text{ and } u_{n-1} = (-1)^{n-2} \frac{(n-2)!}{x^{n-1}}$$

By Leibnitz's theorem, we have

$$(uv)_n = u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \dots + {}^nC_r u_{n-r} v_r + \dots + uv_n$$

$$\begin{aligned} \Rightarrow (x \log x)_n &= (-1)^{n-1} \frac{(n-1)!}{x^n} x + n(-1)^{n-2} \frac{(n-2)!}{x^{n-1}} + 0 \\ &= (-1)^{n-1} \frac{(n-1)!}{x^{n-1}} + n(-1)^{n-2} \frac{(n-2)!}{x^{n-1}} \\ &= (-1)^{n-2} \frac{(n-2)!}{x^{n-1}} [-(n-1) + n] \\ &= (-1)^{n-2} \frac{(n-2)!}{x^{n-1}} \end{aligned}$$

Q.

$$y = e^{2x} \sin x, \quad x \in \mathbb{R}.$$

Use the Leibniz rule to find a simplified expression for $\frac{d^6 y}{dx^6}$.